

When a node reboots, the *Check-In Counter* MAY increase by a larger step than 1. Nodes do not need to write every new value of the Check-In Counter to permanent storage each time it is increased (e.g. to prevent flash wear due to many write operations). One example strategy to achieve reduction of non-volatile storage updates is described below:

1. Read the **counter** value at start-up.
2. Before processing the first message after startup, write **counter + N** to storage, where N is a carefully chosen number (e.g. 1000). This number N should be chosen carefully in order not to exhaust the lifetime 32-bit counter space.
3. Process messages normally until the **counter** has a value one less than the **counter** in storage. When this happens, store **counter + N** to storage.

NOTE

The Check-In Counter has an unlimited lifetime until the device is factory reset. To ensure that a nonce is not reused since it is derived from the Check-In Counter and the ICD key, the key needs to be refreshed before using all valid counter values for the key.

4.6.4. Message Counters as Encryption Nonces

In the context of encrypted messages, message counters serve as nonces for the encryption algorithm, ensuring that every message is encrypted in a unique manner. The uniqueness of an encrypted message's counter is vital to the confidentiality of the message, as the encryption algorithm makes it trivial for an eavesdropper to decrypt messages if the attacker can find two different messages with the same message counter that were encrypted using the same key. Specifically, an attacker can XOR the two different messages that share the same key and nonce to obtain a "block key" which can be used to decrypt any message that uses that key and nonce.

Nodes SHOULD rotate their encryption keys on a regular basis, to ensure that old encryption keys are retired before a *Global Group Encrypted Message Counter* has a chance to wrap to a value previously used with the encryption key. In practice, the frequency of message transmission is such that encryption keys generally rotate at a rate that is much faster than the rate at which a *Global Group Encrypted Message Counter* wraps. In the event that a *Global Group Encrypted Message Counter* wraps before the associated keys are rotated, all keys associated with that *Global Group Encrypted Message Counter* are considered exhausted and are no longer valid to use. In such cases, a new unicast session SHALL be established to the Matter Node to rotate such retired keys before secure communication can resume. Given the importance of confidentiality and message integrity, every effort SHOULD be made to ensure that keys are rotated on a regular basis.

4.6.5. Replay Prevention and Duplicate Message Detection

Beyond their role as encryption nonces, message counters also serve as a means to detect repeated reception of the same message. Message duplication may occur for a number of reasons: out-of-order arrival, network latency, malicious attack, or network error. For example, a duplicate can be caused when a sender retransmits a message after failing to receive an acknowledgement, or because a malicious third party attempted to replay an old message to gain some advantage. To detect duplicate messages, Nodes maintain a history window of the message counters they have received from a particular sender (see [Section 4.6.5.1, "Message Reception State"](#)). Whenever a mes-